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Decentralised Multi-UAV Cooperative Searching Multi-Target in Cluttered and GPS-Denied Environments

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Abstract—In recent years, the use of multi-unmanned aerial vehicles (UAVs) systems in outdoor GPS enabled environments has increased significantly. Their use in GPS-denied environments, however, is still a complex task. Examples of such environments include search and rescue inside buildings or collapsed building scenarios, urban or natural canyons, in which the GPS signal is limited or blocked. In order to provide a possible solution for these applications, this paper presents a multi-UAV system that can be utilised for a team of UAVs to autonomously navigate, search and detect multiple targets in a cluttered and GPS-denied environment. The multi-UAV system uses a decentralised framework based on Decentralised Partially Observable Markov Decision Processes (Dec-POMDP) to formulate the decision-making process considering uncertainties in the environments and sensing. This system was designed for a mission that needs to cooperate efficiently to search and find multiple targets, without knowing possible locations, by sharing limited vital observations through WiFi connection. The system was tested in several cluttered and GPS-denied environments simulated in Gazebo and ROS interface. The environments were populated with different numbers and types of obstacles, several targets, a team of UAVs and boundaries. After comparing three different searching strategies, the results show that the system is capable of cooperatively searching and finding multiple targets in an effective approach while without having a collision in a complex environment. The multi-UAV system is robust to uncertainties in target exploration and localisation, enabling the use of multiple UAVs to enhance critical missions for challenging scenarios such as search and rescue or exploration and disaster management and assessment.

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1. INTRODUCTION

The use of multiple unmanned aerial vehicles (UAVs) for activities such as real-time monitoring, Search and Rescue (SAR), remote sensing, cargo delivery, precision agriculture, and infrastructure inspection has lately received more attention [1–5]. UAVs have been explored for SAR operations due to their manoeuvrability (flying at different speeds, hovering, and agile access to small places), making them ideal for potentially hazardous or challenging tasks for humans [6]. The research on multiple UAVs for SAR missions can be broadly classified into two categories: (i) swarm systems, which consist of a large number of smaller drones with lower-quality sensors and a lower level of autonomy [7], and (ii) cooperative systems, which consist of a few larger UAV platforms with more sensors and more powerful on-board processors that enable more complex on-board decisions than smaller drones [2, 8]. There is a substantial amount of study on SAR missions using a single UAV in GPS-denied settings [9–12]. In comparison, utilising a team of UAVs to efficiently and cooperatively locate a target in congested and GPS-denied environments remains a difficult challenge.

This focuses on the problem of using a team of UAVs for a SAR mission to explore and locate multiple targets within a building structure. This multi-UAV indoor search for multiple targets can be divided into three distinct problems: indoor exploration, multi-UAV cooperation, and multi-target finding. Each UAV must be capable of estimating its own position within the flight area while accommodating all necessary uncertainties regarding its location and motions in the absence of GPS calibration. The multi-UAV system must be efficient and collision-free in order to locate all setup targets while maintaining the system’s scalability and reliability.

The main contributions of this paper are:

1. A scalable decentralised POMDP formulation for simulating a multi-target exploring mission involving a group of UAVs in GPS-denied environments with obstacles in the presence of uncertainty. This formulation enables the management of a variable number of targets without incurring the computational cost associated with an additional target, as the formal POMDP formulation does.
2. This paper proposes method which achieves a combination a assisted localisation function with robust obstacles avoidance function to provide the location of UAV without GPS signals.

2. BACKGROUND

Control Architecture

A multi-UAV system consists of intelligent agents or UAVs as autonomous entities which make decisions according to information of interaction and communication among themselves and with the environment. The control architecture of a multi-agent system (MAS) is defined by interaction within each agent [13]. Most frameworks of a MAS are divided into two major categories: centralised or decentralised architecture [14, 15]. The utilities and requirements of a MAS determine the designed control architecture.

Centralised MAS—A centralised system is composed of a central controller or agent for the whole system and other executive agents. The central controller is capable of communicating and accessing to gain all necessary information or data from other agents in order to process and make decisions for all the agents in the system [14]. In multi-UAV applications, a ground control station (GCS) is the central controller in this multi-UAV system, gathers data from onboard sensors mounted on UAVs, then sends flight commands to all UAVs through communication to complete tasks. It is evident that the centralised one could make an optimised decision from the system level to ensure the efficiency and collaboration of the system when performing tasks. However, the costs of the optimal decision are a stable, unobstructed communication within all the agents in the system and a high demand for computational performance in the central controller. Thus, a centralised MAS has an outstanding performance in limited applications in the real world.

Decentralised MAS—A decentralised system does not have a central controller compared with the centralised one. Instead, there are separate controllers in a MAS for making decisions for one or a group of agents. This provides the benefits for the system, which are less or no constraint on communication, less demanding for computational source and flexibility of scalability [14]. These characteristics prove that the system is more robust and scalable in the number of agents and is able to tolerate communication issues, even in an unconnected circumstance.

MDP and POMDP

Markov Decision Process (MDP) is a widely used mathematical framework to model sequential decision making in stochastic domain [16]. An MDP framework aims to obtain a policy or a strategy from a given reward function, which is designed to filter an optimal action from predetermined actions at a particular decision point depending on detected observation and state of the agent. However, the MDPs framework requires the agent can gain a complete observation at each decision point, which is a demanding restriction for its applications.

Partially Observable Markov Decision Processes (POMDP) extend from MDPs to be able to contain the incomplete observational capabilities of agents and to consider the inaccurate or insufficient observations from sensors and environments as stochastic items in the model. Formally, a POMDP is a tuple described as $\langle S, A, T, \Omega, O, R, \gamma \rangle$ [16], where:

- S is a set of all the environment states for the agent;
- A is the set of all possible actions of the agent;
- $T(s, a, s')$ is the probability of agent in state $s' \in S$ after performs an action $a \in A$ at previous state $s \in S$, which equals with $P_r(s' | s, a)$;

- Ω is the set of all possible observations;
- $O(s', a, o)$ is the probability of observing $o \in \Omega$ at state s' after performed the action a ;
- R is the set of rewards for each chosen action a at every state s ;
- $\gamma \in [0, 1)$ is the discount factor.

The actual state of the agent, known as the belief-state $b(s)$, is described by a probability distribution for all the possible states in agent state-space at the decision time, due to $b(s)$ cannot be directly observable from the tuple. Due to the Markov assumption, it can be updated with a Bayesian filter for every action–observation pair:

$$b'(s') = \eta \Omega(s', a, o) \sum_{s \in S} T(s, a, s') b(s) \quad (1)$$

where η acts as a normalising constant such that b' remains a probability distribution.

The objective of a POMDP is to find a policy that maps beliefs into actions in the form $\pi(b) \rightarrow a$, so that the total expected reward is maximised. This expected reward gathered by following π starting from belief b is called the value function:

$$V(b) = E \left[\sum_{t=0}^h \gamma^t \sum_{s \in S} R(s, \pi(b_t)) b_t(s) | b_0 = b \right] \quad (2)$$

Therefore, the optimal policy π^* is the one that maximises that value function: $\pi^*(b) = \arg \max_a V(b)$.

Multi-Agent POMDP

A multi-agent POMDP framework is an extension from a single agent POMDP framework in order to model the MAS for sequential decision making under uncertainty in a partially stochastic and partially observable environment. Formally, a multi-agent POMDP can be defined as a tuple $\langle I, S, \mathbb{A}, T, \mathbb{O}, O, R, \gamma, h \rangle$ [17], where:

- I is a finite set of n agents, $i \in I$;
- S is a (finite) set of all the environment states for the agents;
- $\mathbb{A}$ is the set of joint actions: $\mathbb{A} = \times_{i \in I} A_i$, A_i is the set of available actions of i -th agent, which can be unique for each agent;
- T is a set of state transition probabilities: $T = S \times \mathbb{A} \times S \rightarrow [0, 1]$, the probability of transitioning from previous state $s \in S$ to current state $s' \in S$ when set of joint actions $\vec{a} \in \mathbb{A}$ are taken by each agent. Hence, $T(s, \vec{a}, s') = P_r(s' | s, \vec{a})$;
- $\mathbb{O}$ is the set of joint observations: $\mathbb{O} = \times_{i \in I} \Omega_i$, Ω_i is a set of observations available to i -th agent;
- O is the observation probability function: $O = \mathbb{O} \times \mathbb{A} \times S \rightarrow [0, 1]$, the probability of seeing the set of joint observations $\vec{o} \in \mathbb{O}$ given the set of joint actions $\vec{a} \in \mathbb{A}$ were taken which results in state $s' \in S$, Hence, $O(\vec{o}, \vec{a}, s') = P_r(\vec{o} | s', \vec{a})$;
- R is the reward function: $R = S \times \mathbb{A} \rightarrow \mathbb{R}$, the immediate reward maps state $s \in S$ and performing the set of joint actions $\vec{a} \in \mathbb{A}$ and is used to specify the goal of the agents;
- $\gamma \in [0, 1)$ is the discount factor for the reward function R ;
- h is the horizon, which the problem terminates at the number of steps after current state.

The goal in a MAS POMDP problem formulation is to calculate an optimal joint policy $\vec{\pi}^* = \{\pi_1, \dots, \pi_n\}$ that maximises the accumulated discounted reward via the reward function (R) as in process in the POMDP model.

The Multi-agent POMDP model can be used for both MAS control architectures, centralised or decentralised. The Decentralised POMDP (Dec-POMDP) is where each agent in a decentralised MAS is to compute its own optimal policy π_i at its local belief b_i with its local observations Ω_i and actions A_i . The Dec-POMDP model has more flexibility of the various grades of communication, from none to perfect and between indirect and direct [17, 18]. The characteristics of the Dec-POMDP model provides a much broader suitable formulation for real-world problems. However, the cost for a Dec-POMDP model to calculate the optimal policy is severely unaffordable for robots, especial for weight limited UAVs. Despite much research that have been done to develop advanced optimal or approximate policy algorithms and simplify the models for specific utilities [17–19], the nature of computational complexity of the model remains NEXP-complete [19]. In addition the complexity of the model surges in an exponentially way when adding an extra agent in the MAS [18, 19]. The intractable computational cost restricts the complexity of the formulation and renders the utilities within simple theoretical cases instead of real-world applications.

3. PROBLEM DESCRIPTION

In this paper, we present a scalable, decentralised multi-UAV system for navigating and exploring finding multiple targets in cluttered environments under UAV position and target uncertainty without access to GPS/GNSS signals. The UAVs in the proposed decentralised MAS use the identical small-size quad-copter platform with onboard downward facing cameras and multi-orientation range sensors. The objective of the system is for UAVs to effectively search and detect all the targets in the search area by using a downward camera on each UAV. The fly areas are illustrated in Figure 1, two/three different industrial-office-like layouts with several walls and obstacles. The number of targets is provided to the system, while the localisations of all targets is uncertain. All the targets are static, placed on the ground and tagged by different ArUco markers for each target. As the typical flight time of a small-size UAV with an onboard computer is in the range of 15 to 20 minutes, a maximum flight time of 720 seconds is specified as a safety margin. The search mission will fail if the UAVs group cannot find all the targets in the area within the maximum flight time.

UAVs are separately deployed from outside of flight areas. All UAVs will take off simultaneously after initialising onboard sensors and communication connections, then hover for a couple of seconds to calibrate the drift error of the IMU, as described in Section 4. The initial drift of localisation in the process of hovering for calibration for each UAV generates an initial uncertainty in the UAV's position and orientation. This initial uncertainty is modelled as a normal probability distribution in the system.

We designed an assistant position subsystem for each UAV to reduce the accumulated errors and drift in IMU over time which has two mechanisms. The first one is that when a UAV flies close to an obstacle, boundary or a wall, the onboard ranger sensor provides a reference distance to that obstacle or boundary to calibrate the position and orientation data from IMU. This mechanism only works once in a period of time

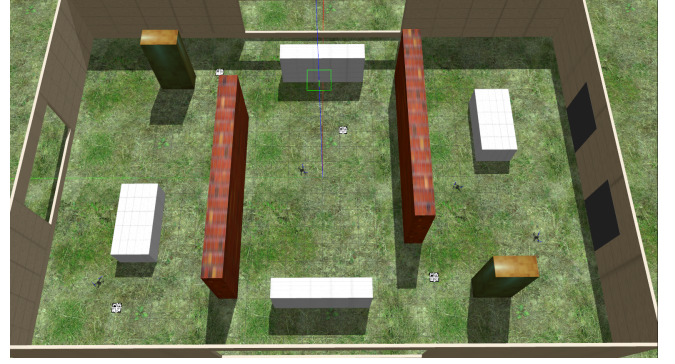


Figure 1. The demonstration of scenario two in simulation environment.

(30 seconds) and helps each UAVs to reduce the accumulated error in localisation. There second mechanism to eliminate the position drift after a period of flight uses the downward camera; once the downward camera of a UAV detects the ArUco marker and the target in its field of view (FOV), an algorithm from the ArUco package, which is based on Mixed Integer Linear Programming for camera to fast and robustly generate estimate pose and location of the marker [20], provides an estimated speed and orientation of the current position drift based on the static target while the flight control subsystem of UAV executes hovering at the exact location. It is assumed that flight conditions remain constant and that there is no wind to disrupt UAV flight performance.

In the SAR mission, the multi-UAV system requires each UAV to be able to avoid collisions with obstacles or other UAVs. The different orientation of the onboard range sensors help the UAV to avoid collisions with obstacles or boundaries, while all the UAVs try to fly at different altitudes in order to avoid collision with each other by broadcasting their altitudes through the communication network. One key feature of the proposed system, is that it can handle an unstable, noisy and disconnected network (e.g. WiFi or radio) which is a common phenomenon in large cluttered environments.

4. SYSTEM ARCHITECTURE

A decentralised and modular multi-UAV framework was developed. The framework can be extended for multiple modular single-UAV in order to ensure the characteristics of Dec-MAS (e.g. scalability and redundancy) [21], as illustrated in Figure 2. Each UAV system is subdivided into four subsystems, each of which contains a variety of modules. Parallel implementation of a single-UAV subsystem in a multi-UAV framework is enabled by the Robotic Operative System (ROS), serving as the interface and connection. These modules communicate via ROS topics that are published and subscribed.

The four subsystems that comprise a UAV framework are as follows: the Mission Planning Subsystem, which determines the optimal flight actions; the Observation Subsystem, which integrates all onboard sensor observations; the Localisation Subsystem, which estimates the position and orientation of the UAVs in the flight space; and the Information Exchange Subsystem, which facilitates the sharing and receiving of observations within the MAS.

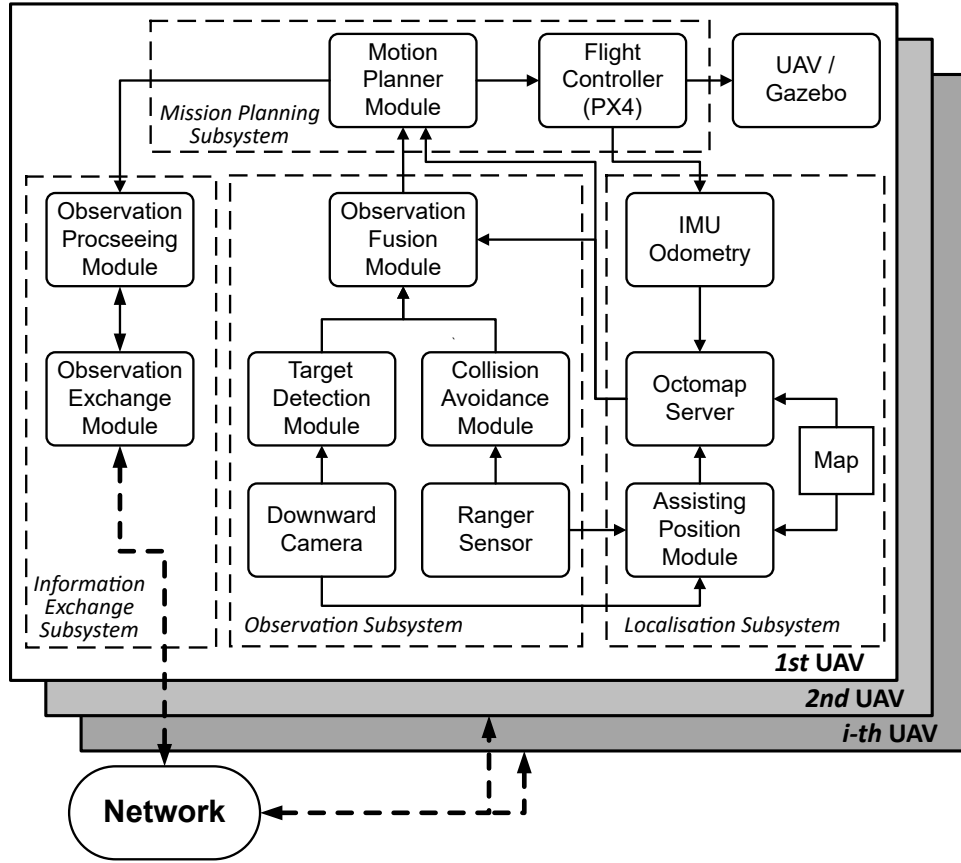


Figure 2. The Int-Dec POMDP Multi-UAV system architecture.

Mission Planning Subsystem

A critical component of an autonomous system is the Mission Planning Subsystem. Its primary function is to determine the next action based on the current optimal policy computed from the decision-making model and currently available information. In the proposed framework, the Mission Planning Subsystem consists of a Motion Planner Module and a Flight Controller. The Motion Planner Module makes use of the TAPIR [22], a fast online solver for large and continuous state space POMDP that makes use of the Adaptive Belief Tree (ABT) algorithm [22], to determine the optimal policy for the decentralised POMDP model (described in Section 5). PX4, an open-source versatile flight control unit (FCU), is a widely used flight controller in autonomous UAV systems [23]. Not only does the PX4 provide PID flight control for a variety of UAV types, including multi-rotor, fixed-wing, and VTOL, but it also includes a high-precision IMU for inertial odometry and a barometer for flight altitude estimation. Additionally, PX4 supports Software In the Loop (SITL) simulation in the Gazebo via ROS as a simulation layer with communication interfaces (MAVLink) between the Motion Planner Module and onboard sensors.

Observation Subsystem

The Observation Subsystem is in charge of all data collected from the environment via various onboard sensors. This subsystem comprises three major modules and their associated sensors: the Observation Fusion Module, the Target Detection Module equipped with a downward camera, and the Collision Avoidance Module equipped with six differently oriented ranger sensors. The Observation Fusion Module

transforms and fuses the data collected by the Observation Fusion Module into the format required by the Motion Planner Module. Additionally, the processed observation is combined with the UAV’s global location and/or the global position of a detecting target as determined by the Localisation Subsystem. To recognise the ArUco markers, a monocular camera configured according to the Arducam IMX477 High Quality motorised focus camera module is downward mounted on the bottom of the UAV. This camera has a resolution of 1440×1080 and a horizontal FOV of 87° . When a marker exposes within the camera’s valid FOV, the Target Detection Module will compute the distance and direction of the detected ArUco marker in the UAV local frame using the ArUco library. Meanwhile, six close-range laser sensors are mounted on the UAV in six directions to ensure collision avoidance during flight. As the typical distance between the rotors of a small size UAV (e.g 80 cm in our experiments), the Collision Avoidance Module is configured to maintain a safe distance (30 cm in our experiments) between the UAV’s rotors and the nearest object by reading data from all the range sensors at a rate of 10 Hz.

Localisation Subsystem

The purpose of this subsystem is to provide an estimated position and orientation of the UAV when it is not receiving GPS signals. It is composed of the IMU Odometry, the Assisting Position Module, and an Octomap Server that contains a known occupancy map of the flying area. The 3D occupancy map is constructed using the environment map in order to visualise the obstacles and boundaries in the probability approach for the Octomap Server [24]. Without GPS signals, the IMU Odometry in PX4 provides a

primary onboard estimated location of the UAV, which has lost accuracy over time due to drift and accumulated error. The Assisting Position Module is intended to mitigate and calibrate IMU drifts and cumulative errors. This module makes use of the precise distance between the UAV and obstacles or boundaries obtained via ranger sensors and the UAV's current possible global position to refer to the given map, compensating for the UAV's accumulated position error throughout the mission. Meanwhile, IMU drift is determined by the horizontal distance between the UAV frame and an ArUco marker. When the Flight Controller believes the UAV is in a stable hover state following takeoff or during any step to detect the tag by a steady hover, the Assisting Position Module computes the actual movement to eliminate the drift error from the IMU. All of these methods reduce the position uncertainty of the UAV in the Octomap Server, and the Octomap Server provides other modules with calculated global location data.

Information Exchange Subsystem

The Information Exchange Subsystem, which is the critical and innovative component of the entire system, consists of two main modules (the Observation Processing Module and the Observation Exchange Module). These two modules extract, simplify, and share observed pivotal information with other UAVs in a designed uniform manner via the communication network at a constant frequency (0.5 Hz), while also receiving and decoding messages (such as the number of detected targets and explored area of the flight space) published by other numbers. We use a WiFi network to connect all of the signal UAV systems in the MAS. The Observation Processing Module is intended to provide a unified approach for processing simple, distinct shared observation data. Each UAV system has a protocol that divides the map of the flight area into a probability grid map with the same granularity as a previously used discrete state space. Each cell in the grid map is assigned a unique identifier (ID) in order to maintain a unified cognition in the multi-UAV system. The Observation Processing Module simplifies the process of sharing observations by checking data from the Motion Planner Module to filter fully complete explored or nearly complete explored grids (over 95% of their area) and sending their IDs to the Observation Exchange Module along with the number of detected targets and their positions on the grid map. The Observation Processing Module updates simplified observations at a rate of (0.5 Hz), the same as the information exchange rate. To avoid UAV-to-UAV collisions, this module constantly monitors the flight altitudes of other UAVs on the network. The module provides an observational input to the Motion Planner Module in order to prevent the UAV from flying at the same altitude as other UAVs.

5. PROBLEM FORMULATION

As described in Section 2, due to the computational complexity of the formal Dec-POMDP and its numerous extensions [18, 19], it is economically impractical for an onboard computer to solve the entire problem, particularly for multi-target search and rescue, even when using state-of-the-art Dec-POMDP solving algorithms. In this paper we propose an integrated Dec-POMDP (Int-Dec-POMDP) model to solve the multiple targets finding problem described in Section 3. The model is based on the formal single agent POMDP which has been extended and modified to solve this task. The Int-Dec-POMDP model is improved and expanded upon from our previous work [5] in order to strike a balance between the computational cost of optimal

Table 1. The Sets of Search and Avoidance Actions

Search Actions a_s	Avoidance Actions a_a
Fly forward	Avoid backward
Fly upward	Avoid upward
Fly downward	Avoid downward
Turn leftward	Avoid leftward
Turn rightward	Avoid rightward
Hovering	

policy computation and the complexity of the formulation implemented in the real world. The Int-Dec-POMDP is a tuple $\langle I, \{S_d, S_t\}, A, T, \{\Omega_l, \Omega_g\}, O, R, \gamma, h \rangle$.

State Variables ($\{S_d, S_t\}$)

I represents the finite set of the identification of UAVs in the multi-agent system. In order to contain the complexity for the multi-target task, we propose that the state of each UAV can be defined with two separate parts:

- the UAV (drone) state $S_d = \langle i, P_r, V_m, d_c, t_f \rangle$ presents the state of the UAV and all the relevant setting parameters for the UAV;
- the target state $S_t = \langle n, P_e, P_t \rangle$ describes the state with targets.

In the UAV state S_d , i is the UAV identification in the multi-UAV system, $P_r = \{x_r, y_r, z_r, \psi_r\}$ are the position and orientation of this UAV in the world frame, $V_m = \{v_x, v_y, v_z, v_\psi\}$ represent the airborne longitudinal, lateral, vertical, and yaw velocities in the UAV frame, safety distance $d_c = \{d_{front}, d_{rear}, d_{left}, d_{right}, d_{up}, d_{down}\}$ are obtained from laser ranger sensors in six directions, and the airborne manoeuvre period in the system is $t_f = 1 \text{ sec}$.

We proposed the target state S_t for maintaining the system state space in the same complexity and being implemented on a onboard computer, instead of using the normal approach of formal POMDP [25]. In the the target state tuple $\langle n, P_e, P_t \rangle$, the number of all the targets is n ; $P_e = \{x_e, y_e\}$ represents the possible position of all targets (not for any specific target) in the process of exploration, which is used to distinguish the unexplored area in the entire space; and $P_t = \{x_t, y_t, z_t, \psi_t\}$ are assigned the actual position and orientation of the target when UAV has detected a target, and this detected target state can represent the localisation of all the targets via Monte Carlo sampling method [26].

Actions (A)

In this formulation, the UAV's flying actions are defined as its longitudinal, lateral, vertical, and yaw movements. The set of actions A is divided into two categories, which are detailed in Table 1: search actions a_s and avoidance actions a_a . To avoid a rapid change in the UAV's flight state, all of the activities summarised in Table 2 are performed at a constant velocity. As a result, the IMU's precision can be maintained for an extended period. The onboard ranger sensors must detect an object inside the specified safety distance before performing one of the avoidance actions. The Motion Planner Module will then select the reverse avoidance action necessary to avoid a collision.

Table 2. The Sets of Search and Avoidance Actions

Action a	Longitudinal Velocity v_x (m/s)	Lateral Velocity v_y (m/s)	Vertical Velocity v_z (m/s)	Yaw Velocity v_ψ ($^\circ$ /s)
Fly forward	1.5	0	0	0
Fly upward	0	0	0.6	0
Fly downward	0	0	-0.6	0
Turn leftward	0	0	0	30
Turn rightward	0	0	0	-30
Hovering	0	0	0	0
Avoid backward	-1.0	0	0	0
Avoid upward	0	0	0.6	0
Avoid downward	0	0	-0.6	0
Avoid leftward	0	1.0	0	0
Avoid rightward	0	-1.0	0	0

Observations ($\{\Omega_l, \Omega_g\}$)

The observations Ω in the system is divided into two components:

Local Observation (Ω_l)—This information is obtained via the local onboard sensors of each UAV, which include the status of target detection in the presence of uncertainty, the position of the detected target within the FOV of the camera, the ID of the explored grid, the status of avoidance mode (in order to maintain the UAV’s safety space from an object), and the distance from surrounding obstacles on the map.

Global Observation (Ω_g)—The global observation from other UAVs is supplied as a shared message, including only the most valuable and simplified observation information via system communication. A shared global observation message contains the following information: the total number of detected targets, the global positions of all detected targets in the grid map corresponding to the detected UAV ID, the most recently explored grid ID corresponding to the searching UAV ID, and the IDs of all explored grids.

Transition Function (T)

The transition function (T) models the UAV dynamic response to motion commands that are represented as the step inputs. In order to represent the i -th UAV dynamics, a kinematic model is described as follows:

$$P_r(t+1) = P_r(t) + T_r(t) \Delta P_r(t) \quad (3)$$

$$\Delta P_r(t) = V_m(t) \Delta t + P_{wt}(t) \quad (4)$$

In the above equations, $T_r(t)$ is the dynamic change matrix at time step t , $P_r(t)$ is the position of i -th UAV in the world frame at time step t and $P_{wt}(t) = \{x_{wt}(t), y_{wt}(t), 0, \psi_{wt}(t)\}$ is the pose uncertainty during the flight. The position of each UAV is determined by calculating the change in position due to its current velocity (Equation 3), which is controlled by the motion planner module and is selected according to the commanded action. A transformation from the UAV’s frame to the world frame is also calculated in these equations [27].

Reward Function (R)

The reward or cost values in the reward function are used to tune the solver in the system in order to force the UAV to follow the pattern of actions proposed. At each step, the

Table 3. Reward Function Sets

Status & Actions	Reward/Cost Value		
	Set A	Set B	Set C
Detecting a target	300	300	300
Exploring a grid	100	100	100
Hitting an obstacle	-100	-300	-300
Out of region	-100	-300	-300
Search actions	-10	-10	-10
Hovering	-30	-30	-30
Avoidance actions	-10	-10	50

reward function is composed of the following components: (i) the reward for locating the target and exploring a new grid on the map, (ii) the penalty for colliding with an obstacle or flying outside of operational space, and (iii) the cost of each manoeuvre. Three value sets of the reward function are listed in Table 3 to compare the multi-UAV system’s navigation and obstacle avoidance capabilities. These value sets include different penalties for colliding with an obstacle, a boundary or other UAVs and different costs for an avoidance action. Due to the fact that hovering reduces the efficiency of a SAR mission, the cost of hovering is higher than the cost of other search actions.

Uncertainty

The system’s three primary sources of uncertainty are the localisation uncertainty of the UAVs during flight, the location uncertainty of the targets, and the uncertainty associated with shared observations via the communication.

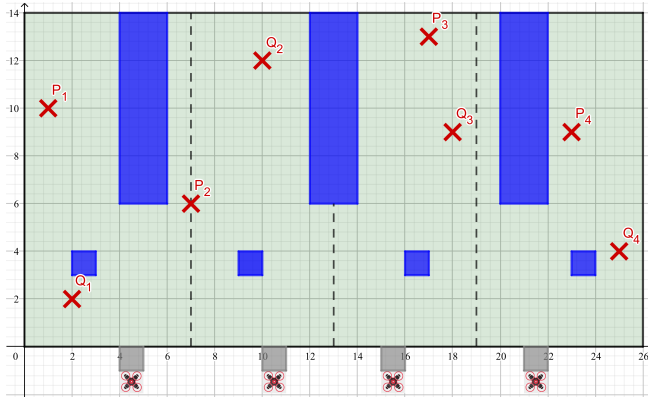
Localisation of UAVs—The uncertainty in the location of each UAV in this formulation consists of two parts: (i) the uncertainty in localisation of system once it is initialised (on the $x - y$ plane) is $P_{wd} = \{x_{wd}, y_{wd}, 0, \psi_{wd}\}$, in the process of initialisation, represented by a Gaussian probability distribution ($\mu = 0.0m, \sigma = 0.2m$); (ii) the UAV’s pose uncertainty after executing an action: $P_{wt} = \{x_{wt}, y_{wt}, 0, \psi_{wt}\}$, which is also described by the Gaussian distribution ($\mu = 0.0m, \sigma = 0.2m$) & ($\mu = 0.0^\circ, \sigma = 5.0^\circ$). We assume all the altitude data from PX4 is accurate for the system through the flight. These localisation uncertainty values are updated via the transition function.

Position of Targets—The position uncertainty of the target $S_t = \{P_e, P_t\}$ are represented by the same uniform distribution in the formulation.

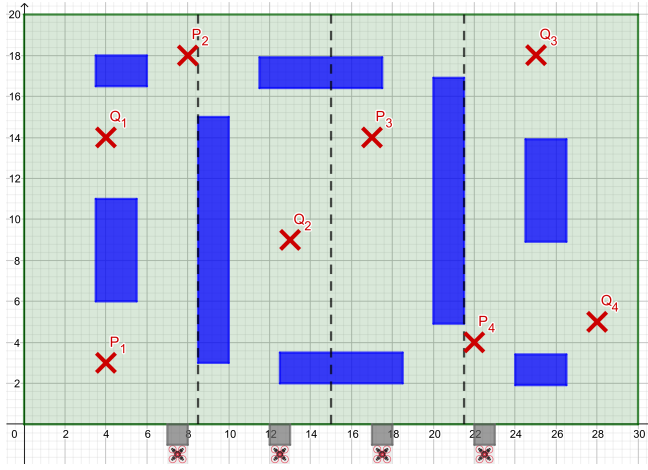
Exchanged Observation—Since the threshold for an explored cell in the grid map is 95% (as mentioned in Section 4), there is some uncertainty in shared observations regarding explored grids. We use a sample filter to retain the 5% of state samples within the informed grid as the uncertainty of the exchanged observation.

6. RESULTS

Three types of experiments were conducted to evaluate and analyse the proposed multi-UAV system’s performance. We use a simulation environment comprised of ROS, Gazebo, PX4 SITL, and Rviz. Figure 3 illustrates the two scenarios (S1 & S2) used in the simulations. Each simulated scenario contains two different sets of target locations (denoted by red



(a) Scenario one (S1), with target locations P and Q.



(b) Scenario two (S2), with target locations P and Q.

Figure 3. Simulation environments with two unique scenarios.

crosses in the figure and by ArUco markers in the Gazebo simulation). A team of UAVs take off from the grey cells at the bottom of the image and fly through the green fly area in search of all targets while avoiding collisions with the blue obstacles. To ensure precise detection of the ArUco marker, each UAV's maximum flight altitude was limited to 4.5 m.

Experiment One

The purpose of this experiment is to compare the effect of various combinations of reward/cost values. The collision rate results are summarised in Table 4, which includes 25 runs for each scenario using four UAVs to locate targets at position P. The collision rate indicates how many times all UAVs in the team collided with an obstacle, a boundary wall, or another UAV during a single mission. It is evident that scenario two is more cluttered than scenario one, as indicated by the collision rates of each reward set in Table 4. When compared to Set A, the collision rates of reward values Set B and Set C are significantly lower. This demonstrates that increasing the penalty for a collision in the reward function contributes to the creation of a safe path when a UAV performs a task. The distinction between Set B and Set C is that the avoidance actions are either rewards or costs in the reward function, and the close-to-zero results of Set C demonstrate the utility and robustness of the collision avoidance function.

Table 4. Summary of Collision Rate

Reward Function	Scenario	Collision Rate
Set A	S1	36 %
Set A	S2	44 %
Set B	S1	8 %
Set B	S2	12 %
Set C	S1	0 %
Set C	S2	4 %

Table 5. Summary of Success Rate of All Targets Detected

Target Position Set	Scenario	Success Rate
P	S1	90%
Q	S1	95%
P + Q	S1	80%
P	S2	70%
Q	S2	75%
P + Q	S2	55%

Experiment Two

The mission success rate and flight time were compared in both scenarios using three target position sets. Each case consisted of twenty runs using a four-UAV system for reward Set C. Table 5 compares target detection rates for all cases. Overall, the results of scenario one are superior to those of scenario two. As can be seen, scenario two's layout is significantly larger and more complex than scenario one's, and scenario two requires UAVs to spend more time exploring to locate all targets. In both scenarios, the UAV team's success rate in detecting eight targets in position P & Q is significantly lower than in detecting four targets in positions P or Q.

The Figure 4 illustrates the time required to locate all setting targets in both scenarios. Overall, it is clear that the proposed multi-UAV system took longer to detect all targets in scenario one than it did in scenario two, owing to the complexity and size difference of the test environments. The primary differences between the results for finding four targets (located at P or Q) and eight targets (located at P & Q) in both scenarios are that the median time required to find more targets was significantly longer, and the interquartile range (IQR) was also noticeably wider. This is expected, as the area explored to locate eight targets is more extensive than the area explored to locate four targets. Meanwhile, the set positions for the same number of targets had a moderate effect on the results in various scenarios.

Experiment Three

The purpose of this experiment is to determine the effect of the team size on the performance metric (time to find all targets) of the proposed multi-UAV system. This case study was repeated 15 times with four different groups of UAVs in order to locate all targets placed at location P in scenario one. The number of UAVs in each of these four groups was increased from two to five to demonstrate the multi-UAV system's scalability. According to the results shown in Figure 5, as the number of UAVs increases, the average time required to find all targets decreases with a decreasing

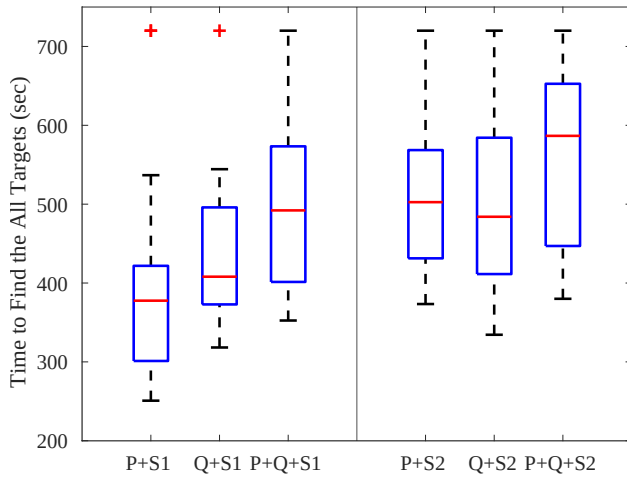


Figure 4. Time to find all targets for both scenarios for three sets of target locations.

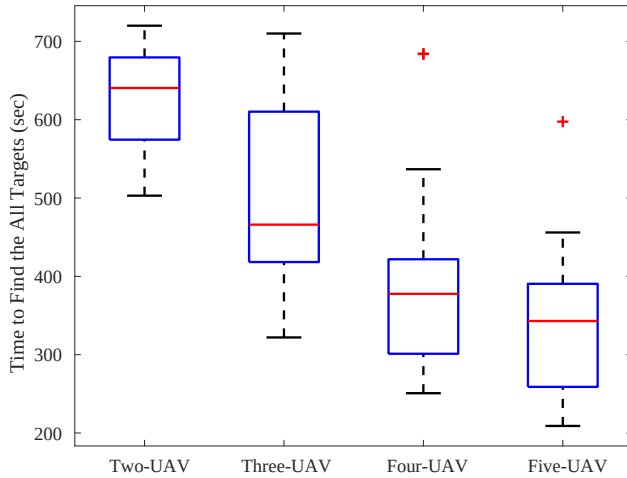


Figure 5. Time of different numbers of UAVs in the team to find four targets located at P in scenario one (S1).

trend, from 640.5 *sec* to 342.9 *sec*. As can be seen, increasing the number of UAVs in a two-UAV system increases system performance by 37.4 % while decreasing the average search time by 10.2 % when comparing a four-UAV system to a five-UAV system under the same simulated condition. This decrease in improvement in results indicates that increasing the number of UAVs does not improve the performance of the multi-UAV system linearly and becomes marginal as the number of UAVs increases. This trend creates the issue of having too many UAVs in the system, which increases the danger of collision, energy consumption, and system cost significantly, while performance and efficiency improve just marginally.

7. CONCLUSIONS & FUTURE WORK

This article described an approach and framework for multi-UAV system search in an indoor environment without the use of GPS/GNSS. By utilising the proposed Int-Dec-POMDP as the underlying model, the framework is able to account for formulation uncertainties while maintaining the benefits of a multi-agent system. The extended observation and observation exchange subsystems are essential and new features that

enable the system to maintain scalability and collaboration while containing computing complexity.

Additionally, this paper proposes a combined approach for assisting navigation and localisation in the absence of GPS/GNSS signals, as well as for robust collision avoidance. Moreover, a contrast test of different sets of reward values in the reward function is simulated to demonstrate the collision avoidance function's influence and robustness.

The paper's primary contribution is a demonstration of how the proposed framework may be used to perform MAS for multi-target search in a GPS-denied environment in an efficient and collision-free manner. The computation cost of the proposed multi-UAV system is within the capabilities of onboard processors, which is not possible when using formal Dec-POMDP formulations.

Additional research avenues include identifying and integrating a cost-effective Simultaneous Localisation and Mapping (SLAM) system within the architecture of a multi-UAV system. Additionally, real-world experiments can be conducted to demonstrate the framework's adaptability by examining other limitations on communication network systems.

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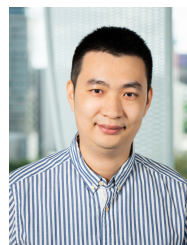
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